

Final Executive Report

End-to-End Business Analytics: Retail Sales

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Tools used: Excel, SQL and Power BI

1. Executive Summary

The business sold €1,552,071 across 11,971 orders between January 2022 and January 2025. The average order is worth €129.65, and 25 customers placed all of them.

Performance is steady and evenly spread. Sales fell 3.7% in 2023, then rose 6.8% in 2024 to €524,881, the best year on record. Across the two years from 2022 to 2024, sales grew only 2.9%. The business is stable but not growing.

No category, region or customer stands out as a risk. The top category leads the bottom one by 13.5%, and the top region leads the bottom one by 3.1%. Growth has to come from smaller levers: a higher value per order, a plan built around the January peak, and cleaner data. The five recommendations in this report target those levers. Each one that can be measured is worth between 1.0% and 1.4% of sales.

Total Sales	Total Orders	Total Customers	Average Order Value	2024 vs 2023
€1,552,071	11,971	25	€129.65	+6.8%

2. Business Problem and Data

Problem. Management has several months of sales records but no clear view of how the business is doing. They need to know what sells, where, to whom and when, and what to do about it.

Dataset. The source is the public Kaggle dataset Retail Store Sales: Dirty for Data Cleaning, with 12,575 raw rows. Each row is one order with a customer, category, item, price, quantity, total, payment method, channel, date and discount flag. After cleaning, 11,971 orders remain.

Tables. For SQL and Power BI the data is split into Orders, Customers, Products, Categories, Regions and Calendar. Orders is the fact table and the rest are dimensions.

Metrics. Total sales, total orders, total customers, average order value, sales growth and each category's share of sales.

3. Business Questions

#	Question	Answer
1	What is the overall sales performance?	€1,552,071 from 11,971 orders at €129.65 each. Annual sales were €510,330 in 2022, €491,312 in 2023 and €524,881 in 2024.
2	Which category earns the most?	Butchers earns the most at €208,118. Milk Products earns the least at €180,112.
3	Which region performs best?	Central leads at €314,780. West is lowest at €304,927, a gap of 3.1%.
4	Who are the most valuable customers?	CUST_24 at €68,452, CUST_08 at €67,352 and CUST_05 at €66,975. The top customer spends only 1.2 times the lowest.
5	How do sales change over time?	Sales are flat overall. January is the peak, February falls 19.6% after it, and 2024 rose 6.8%.

#	Question	Answer
6	Which products or regions underperform?	West is the lowest region and Milk Products the lowest category. The lowest named products are Item_3_Ehe at €224 and Item_5_Bev at €517.
7	Do discounts change how much customers buy?	No. Discounted orders average 5.53 units and €130.49. Full-price orders average 5.54 units and €129.23.
8	What should management do?	Raise the value of each order, plan around the January peak, test discounts and fix item data. Section 7 gives the detail.

4. Analysis Summary by Tool

Excel: preparation and exploration

The raw file held 12,575 rows. 604 rows had no quantity and no total, so nothing could rebuild them and they were removed. In the remaining rows, missing prices, quantities and totals were rebuilt from the other two fields. 609 orders had no item name and were labelled Unknown Item. Blank discount flags were set to FALSE, text was trimmed and standardised, and dates were converted to true dates. No duplicate records were found.

Formulas covered SUMIFS, COUNTIFS, AVERAGEIFS, IFS and lookups. Four pivot tables covered category by region, customer segment, month and quarter. The main patterns were the narrow gaps between categories and regions, the January peak, and the 2024 recovery.

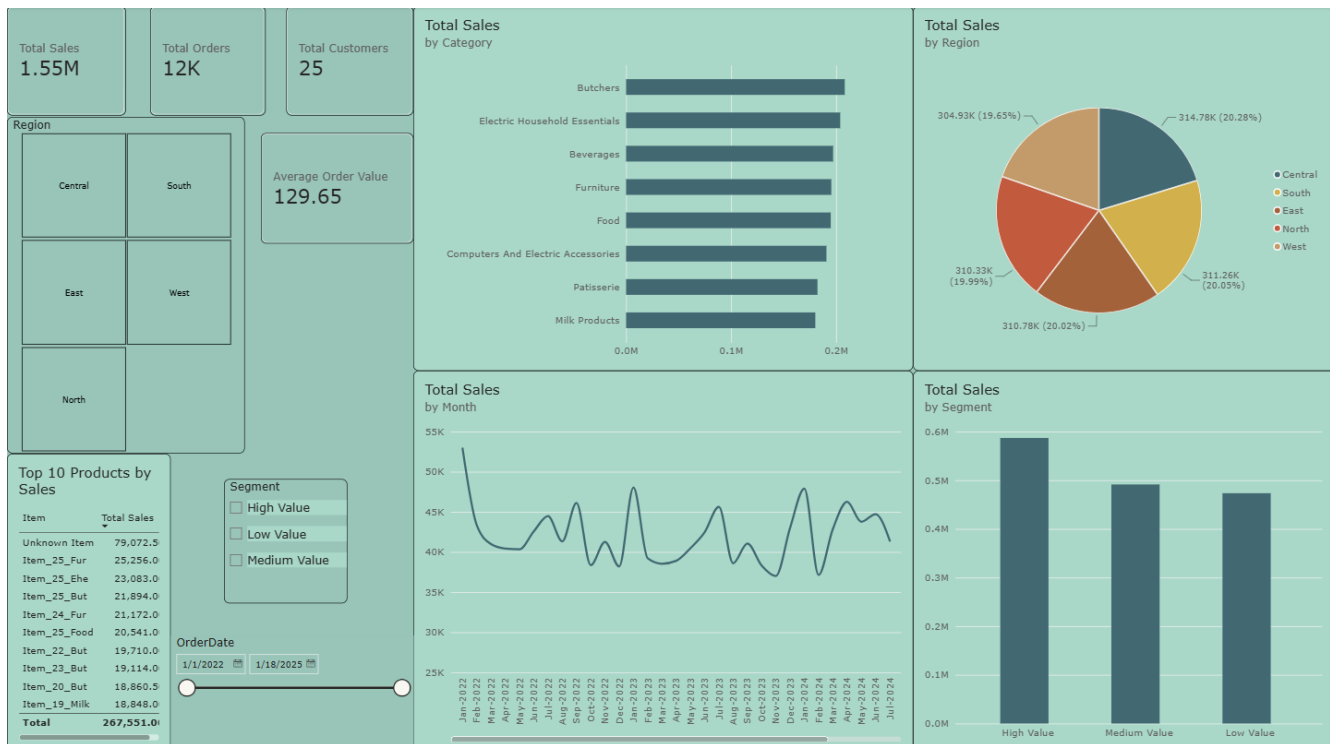
SQL: business questions

The data sits in a SQLite database with Orders as the fact table and Customers, Products, Categories and Regions as dimensions. Queries covered filtering, grouping with HAVING, and inner and left joins. A left join confirmed that every customer has placed orders. Products are keyed on item and category together, because Unknown Item appears in all eight categories and would otherwise merge them.

Power BI: model and dashboard

Orders connects to Customers, Products and Calendar with one-to-many relationships. The measures cover total sales, total orders, total customers, average order value, sales growth and category share. Slicers for region, segment and date range filter every visual on the page.

5. Dashboard



Executive dashboard in Power BI. Each visual answers one management question.

Visual	Question it answers
KPI cards	How big is the business and what is a typical order worth?
Sales by Category	Which product groups earn the most?
Sales by Region	Where does revenue come from?
Sales by Month	How do sales move over time?
Sales by Segment	Which customer group contributes the most?
Top 10 Products	Which items earn the most?
Slicers	How do the answers change for one region, segment or period?

6. Business Insights

Five Key Findings

- 1. Sales are stable, not growing.** Annual sales were €510,330, €491,312 and €524,881 for 2022, 2023 and 2024. The net change over the period is +2.9%.
- 2. Categories differ mainly by order value.** Butchers leads at €208,118 and Milk Products trails at €180,112, a gap of 13.5%. Order counts are almost equal, from 1,441 to 1,525 per category. Butchers averages €139.12 per order and Milk Products €119.04.
- 3. Regions are close.** Central leads at €314,780 and West trails at €304,927. Each region has between 2,361 and 2,434 orders. East has the most orders but the lowest average order at €127.68.
- 4. Customers are evenly matched.** Each of the 25 customers placed between 446 and 519 orders and spent between €57,156 and €68,452. The High Value segment of 9 customers brings in 37.8% of sales, close to its 36% share of customers. Revenue does not depend on a few accounts.
- 5. No product dominates.** The best named product, Item_25_Fur, earns €25,256, or 1.6% of sales. The top 10 named products earn 13.3%. Unknown Item, a data gap, would rank first at €79,072.50.

Three Important Trends

- 1. January is the peak month.** January averages €49,624, which is 17% above the average month of €42,403. It was the strongest month in 2022 and 2023 and the second strongest in 2024, behind December. February then falls 19.6% from January.
- 2. The 2024 recovery came from two quarters.** Compared with 2023, Q2 rose 10.6% and Q4 rose 14.3%. Q1 rose 1.7% and Q3 rose 1.3%. Q4 2024 at €135,279 is the best quarter since Q1 2022.
- 3. Revenue follows price, not demand.** An item's price has almost no link to how often it is ordered, with a correlation of 0.06. Its link to revenue is strong, at 0.65. Items above €30 make 30.4% of orders and 46.8% of sales. Items at €15 or less make 25.2% of orders and 10.2% of sales.

Three Problems and Opportunities

- 1. Growth is flat.** Two years of movement add up to +2.9%, and no lever drives steady growth.
- 2. Small orders in Milk Products and Patisserie.** Their average orders are €119.04 and €126.42, below the company average of €129.65, while their order counts are normal.
- 3. Discounts and item data.** One in three orders carries a discount with no lift in basket size, and 5.1% of sales have no item recorded.

7. Management Recommendations

Recommendation 1: Move customers up a price tier

Finding	Customers order low-price and high-price items about equally often, but items at €15 or less bring in 10.2% of sales from 25.2% of orders.
Business Impact	Revenue depends on the price of the item chosen. A quarter of orders bring in a tenth of sales.
Recommendation	Pair low-price items with a mid or high-price item from the same category, using bundles and add-on prompts at checkout.
Expected Outcome	If 10% of low-price orders, 302 orders, move from the €52.37 average to the €125.31 mid-price average, sales rise by $302 \times €72.94$, about €22,000 over the period, or 1.4%.

Recommendation 2: Lift order value in Milk Products and Patisserie

Finding	Milk Products has 1,513 orders, above the category average of 1,496, yet the lowest sales at €180,112. Each order averages €119.04. Patisserie averages €126.42.
Business Impact	These categories lose sales through small orders, not weak demand.
Recommendation	Introduce multi-buy offers and larger pack sizes in both categories.
Expected Outcome	Bringing Milk Products to the €129.65 company average adds $1,513 \times €10.61$, about €16,050, or 1.0% of sales. Patisserie adds $1,441 \times €3.24$, about €4,660.

Recommendation 3: Plan around the January peak

Finding	January averages €49,624, 17% above the average month, and February falls 19.6% right after. February and October are the weakest months at about €39,900 and €39,800, roughly 6% below average.
Business Impact	Stock and staffing set for an average month run short in January and sit idle in February and October.
Recommendation	Build stock and staffing plans for the January peak, and schedule promotions in February and October.
Expected Outcome	Lifting February and October to the average month of €42,403 adds €2,508 plus €2,599, about €5,100 a year, or 1.0% of a year's sales.

Recommendation 4: Test whether discounts work

Finding	4,019 of 11,971 orders, 33.6%, carry a discount. Discounted orders average 5.53 units and €130.49. Full-price orders average 5.54 units and €129.23.
Business Impact	A discount that does not enlarge the order costs margin for no return. The data does not record the discount amount, so the cost is not yet measurable.
Recommendation	Record the discount value on every order. Then withhold discounts in one region for a quarter and compare order counts and order value with the other regions.
Expected Outcome	A clear answer on whether discounts add orders. If they do not, the discount cost on one in three orders can be cut.

Recommendation 5: Fix item capture at order entry

Finding	609 orders, 5.1% of orders and €79,072.50 of sales, have no item recorded. Unknown Item would rank as the top product.
Business Impact	Product decisions ignore 5.1% of sales and the top-product ranking is wrong.
Recommendation	Make item ID a required field at order entry and recover missing items from source system logs.
Expected Outcome	Every euro of sales is tied to a product, and the product ranking becomes reliable.

8. Limits of the Data

The source data has no geographic field. Each customer was assigned to one of five regions with a lookup table, so regional results depend on that assignment.

Customer segments split the 25 customers into thirds by total spend. Spend is close across customers, so the segments differ by little.

The data ends on 18 January 2025, so January 2025 is a partial month and is left out of yearly comparisons. The data holds no cost or margin, so profit is not measured.

9. Conclusion

The business is stable and evenly spread across categories, regions and customers, and it has been flat for three years. Nothing is failing and nothing is pulling ahead.

The best gains sit in how much each order is worth, when demand peaks and how clean the data is. The five recommendations target those areas, and each can be tested with the data the business already collects.